Mohammad Karimi

Staff AI Engineer at Proofpoint

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Summary

- Staff AI engineer who architects and ships **multi-agent**, **RAG**, and **document-intelligence** systems end to end, from research prototype to Kubernetes services handling **20M+ messages/day**
- Expert in cost-efficient LLM inference (**vLLM**, quantization, continuous batching, KV caching), repeatedly delivering **10–122×** cost reductions over managed AWS services
- Built natural-language systems over regulated financial and compliance data on **AWS, GCP, and Azure**; published NLP researcher and advanced ML/MLOps instructor

Experience

Staff AI Engineer Jul. 2025 – Present

Proofpoint, Toronto, ON

- Architected Proofpoint Automate's production LLM inference platform on **Kubernetes** (**20M+ messages/day**), serving **20+ open-weight models** via **vLLM** with AWQ quantization, prefix/KV caching, and continuous batching at **10–15× lower cost** than AWS Bedrock
- Built a **sub-second speech-to-speech multi-agent system** for Compliance Investigator, designing a unified context layer over **knowledge graphs**, entity resolution, and parallel **LLM subagents** for real-time investigative Q&A
- Engineered a cascaded **agentic decisioning pipeline** for Proofpoint Supervision that routes traffic across classical ML filters, tool-calling LLMs, and **human-in-the-loop (HITL)** active learning, cutting false-positive escalations by **45%**
- Distilled frontier model reasoning into a **QLoRA-tuned open-weight LLM** that beat commercial compliance baselines on golden evals; boosted AWS Nova F1 by **+40%** via structured function calling and **DSPy/GEPA** prompt optimization
- Designed a petabyte-scale **hybrid RAG** search engine over millions of archived emails combining dense vector embeddings, sparse **BM25**, and a **cross-encoder re-ranker**, achieving **94% top-5 recall** and cutting query latency by **65%**
- Fine-tuned **Longformer** in PyTorch for regulated long-document classification (**90%+ F1**) with token-level **Integrated Gradients** explainability for audit trails, alongside a hybrid **BERT** PII Redaction API (**122× cheaper** than AWS Guardrails)

Senior Data Scientist Jun. 2023 – Jul. 2025

KPMG Lighthouse, Toronto, ON

- Migrated **Prudential Financial** from an on-prem stack to **GCP** and integrated **agentic LLMs** with **RAG** for context-aware database queries, improving scalability, security, and cost
- Built a generative SOC report system on GCP with instruction-tuned local **LLMs** and accelerated inference, reaching **92%+** accuracy on information retrieval from unstructured reports
- Contributed to **nine production pipelines** for Bell on **Vertex AI** (customer churn), from ingestion through batch deployment, orchestrated with **Cloud Composer**, Pub/Sub, and BigQuery

Associate Machine Learning Developer Sep. 2022 – May 2023

AltaML, Edmonton, AB

- Shipped a production bias and hate-speech classifier on **Azure** using **GPT-3.5**, covering multiple bias types in social-media text
- Built a scalable Azure ML pipeline for the Government of Alberta that scored **2.6B+** geospatial points at **88%+** accuracy under cross-validation

Data Scientist and NLP Researcher Jan. 2021 – Aug. 2022
University of Alberta, Edmonton, AB

- Fine-tuned neural language models to detect ungrammatical writing in large learner corpora at **97%+** accuracy; results published in the *International Journal of Artificial Intelligence in Education* (2025)

Software Engineer Sep. 2018 – Jan. 2020
Iranian National Robotics Team, Tehran, Iran

- Built an imagery acquisition pipeline and a **Faster R-CNN** detector for object recognition

Education

M.Sc. in Computer Science Jan. 2021 – Aug. 2022
University of Alberta *Edmonton, AB*

B.Sc. in Software Engineering Sep. 2015 – Sep. 2019
University of Science and Technology of Mazandaran *Mazandaran, Iran*

Skills

Languages: Python, SQL, C++, Java, JavaScript
ML & GenAI: PyTorch, Hugging Face, vLLM, LangChain, LangGraph, DSPy, Unsloth, QLoRA, scikit-learn, XGBoost, spaCy
MLOps: Kubernetes, Docker, AWS, GCP, Azure, Airflow, Cloud Composer, Vertex AI, BigQuery, Git, Linux
Data: PostgreSQL, MySQL, SQL Server, SQLite
Certifications: GCP Professional Machine Learning Engineer; Azure AI Engineer; 20+ additional cloud certifications

Publication & Mentorship

- **Karimi Abdolmaleki, M.**, Farias Wanderley, L., Cutumisu, M., et al. Negative Language Transfer Identification in the English Writing of Chinese and Farsi Native Speakers. *Int. J. Artificial Intelligence in Education* (2025). doi.org/10.1007/s40593-025-00468-8
- **ML Instructor, Alberta AI Association.** Taught an advanced machine learning course to 15+ students, covering NLP and MLOps from data processing through training, evaluation, and deployment